



NUS
National University
of Singapore

EE5111/EE5060/EE5061

**Selected Topics in
Industrial Control & Instrumentation**

National University of Singapore

EE5111

Selected Topics in Industrial Control & Instrumentation

EE5060

Sensors and Instrumentation for Automation

EE5061

Industrial Control and Programming

WEEK 1 to WEEK 6	EE5060 Sensors and Instrumentation for Automation	EE5111
RECESS WEEK		
WEEK 7 to WEEK 13	EE5061 Industrial Control and Programming	

EE5111/EE5060/EE5061

Target students:

MSc, Meng/PhD

Pre-Requisite:



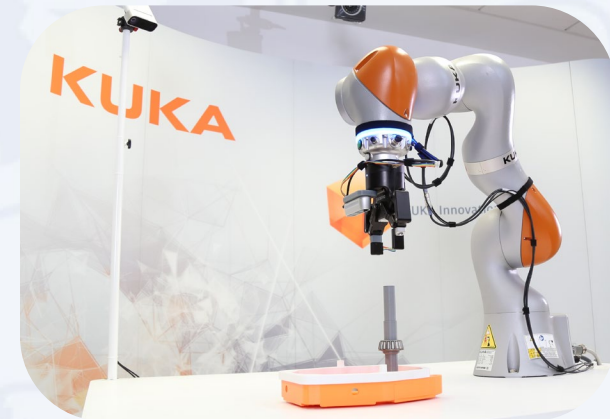
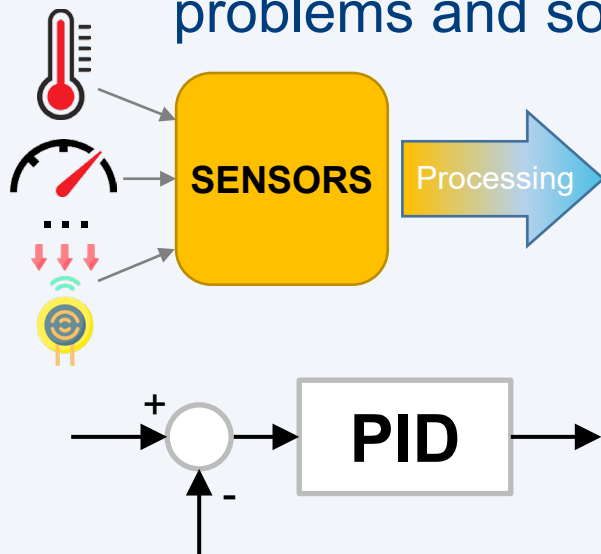
Background in **feedback control systems or relevant experience**

Preclusions:

**EE5060, EE5061 (for EE5111)
EE5111 (for EE5060 & EE5061)**

Module Description

- The module offers students timely and updated coverage of a wide range of topics relevant to **common industrial practice and control, smart sensor and instrumentation** tapping on the latest and diverse range of developments in the repertoire of the control group and collaborating companies and institution
- The nature of the module allows the flexibility for recent topics, problems and solutions to be shared with the students



Lecturers

- **Lin Feng, Dr**
– Senior Lecturer, NUS
feng_lin@nus.edu.sg
- **Jiang Rui, Dr**
– Adjunct Lecturer, NUS
elejiangrui@nus.edu.sg
- **Murali Krishnan Thiagarajan, Mr**
– Adjunct Senior Lecturer, NUS
elemkt@nus.edu.sg
- **Huang Sunan, Dr**
– Senior Research Scientist, Temasek Laboratories@NUS
tslhs@nus.edu.sg
- **Liang Wenyu, Dr**
– Adjunct Assistant Professor, NUS
liangwenyu@nus.edu.sg

Teaching Assistants

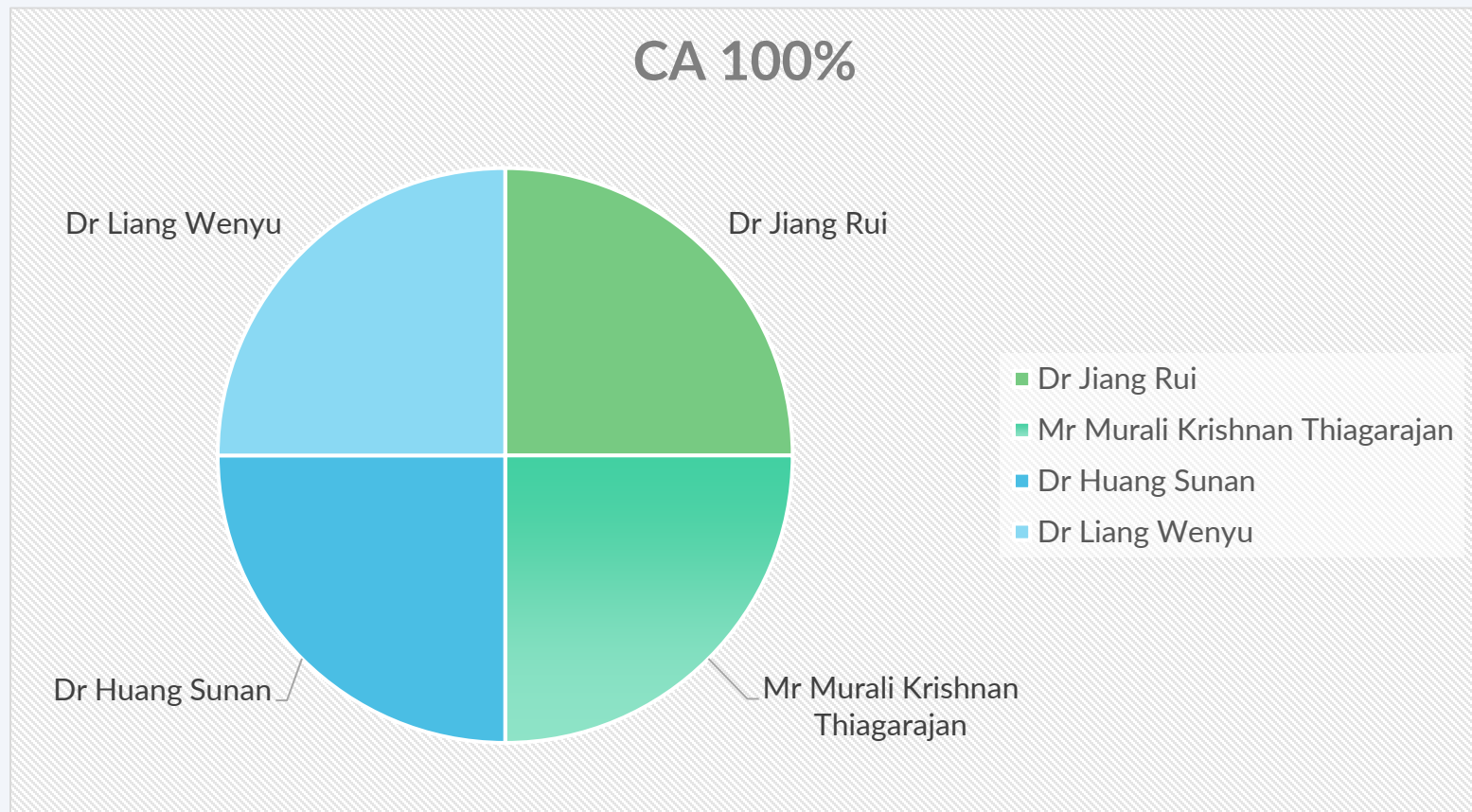
- **Xue Junyuan, Mr**
xue.junyuan@u.nus.edu
- **Xu Yilan, Ms**
xuyilan@u.nus.edu

Schedule

WEEK 1	Basics of Sensors and Instrumentation (Dr Jiang Rui)	Aug. 12, 2026
WEEK 2	Vision Sensors (Dr Jiang Rui)	Aug. 19, 2026
WEEK 3	Fieldbus, Networking and Web Server (Mr Murali Krishnan Thiagarajan)	Aug. 26, 2026
WEEK 4	Sensor Fusion (Dr Jiang Rui)	Sep. 02, 2026
WEEK 5	Sensors, Wireless and IIoT (Mr Murali Krishnan Thiagarajan)	Sep. 09, 2026
WEEK 6	Force and Tactile Sensing (Dr Liang Wenyu)	Sep. 16, 2026
RECESS WEEK		
WEEK 7	Industrial Application (Mr Murali Krishnan Thiagarajan)	Sep. 30, 2026
WEEK 8	Fault Diagnosis and Fault Tolerance Control I (Dr Huang Sunan)	Oct. 07, 2026
WEEK 9	Fault Diagnosis and Fault Tolerance Control II (Dr Huang Sunan)	Oct. 14, 2026
WEEK 10	Fault Diagnosis and Fault Tolerance Control III (Dr Huang Sunan)	Oct. 21, 2026
WEEK 11	Advanced PID Control and Tuning (Dr Liang Wenyu)	Oct. 28, 2026
WEEK 12	Precision Motion Systems (Dr Liang Wenyu)	Nov. 04, 2026
WEEK 13	Simulation Using ROS (Dr Lin Feng)	Nov. 11, 2026

Assessment

- **100% CA**
 - mainly from assignments, or project reports



Assessment

WEEK			2026 Sem 1	% of CA Marks
1	EE5111 (EE5060)	Basics of Sensors and Instrumentation	Dr Jiang Rui	25% (CA1) THREE <u>in-class</u> quizzes
2		Vision Sensors		
3		Sensor Fusion		
4		Fieldbus, Networking and Web Server	Mr Murali Krishnan Thiagarajan	20% (CA2)
5		Sensors, Wireless and IIoT		
6		Force and Tactile Sensing	Dr Liang Wenyu	5% (CA3)
RECESS				
7	EE5111 (EE5061)	Industrial Application	Mr Murali Krishnan Thiagarajan	5% (CA4)
8		Fault Diagnosis and Fault Tolerance Control I	Dr Huang Sunan	25% (CA5)
9		Fault Diagnosis and Fault Tolerance Control II		
10		Fault Diagnosis and Fault Tolerance Control III		
11		Advanced PID Control and Tuning	Dr Liang Wenyu	20% (CA6)
12		Precision Motion Systems		
13		Simulation Using ROS	Dr Lin Feng	0%

NUS
National University
of Singapore

[illegible]